

Lightweight High-Voltage Power Converters for Electroaerodynamic Propulsion

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Abstract—Recent studies in electroaerodynamic (EAD) propulsion have stimulated the need for lightweight power converters providing outputs at tens of kilovolts and hundreds of watts [2]–[4]. This article demonstrates a design of a lightweight high-voltage converter operating from a 160–225 V dc input and providing dc output of up to 565 W at 40 kV. It operates at around 500 kHz and achieves a specific power of $1.15 \text{ kW} \cdot \text{kg}^{-1}$. High-voltage converters generally comprise an inverter, a step-up transformer, and a rectifier, with the large needed voltage gain distributed among these stages. Several means of realizing these stages are compared in terms of weight. The weight of the converter is minimized by properly selecting and optimizing the design and the voltage gain of each stage within the constraints of device limitations and losses. A prototype circuit is developed based on this approach and used to drive an EAD propulsion system for an unmanned aerial vehicle. Moreover, this article also presents approaches to further improve the specific power of such converters, including better diode utilization and more flexible high-voltage transformer design. In addition to addressing the needs for EAD, this research can potentially benefit the development of lightweight high-voltage converters in many other applications, where weight and size are important.

Index Terms—Cockcroft–Walton, Dickson, electroaerodynamic (EAD), high-voltage converter, high-voltage diodes, lightweight.

I. INTRODUCTION

HIGH-VOLTAGE dc–dc power converters are essential in many industrial, medical, and aerospace applications [5]–[31]. The specific power, defined as power delivered per unit of weight ($\text{kW} \cdot \text{kg}^{-1}$), of high-voltage dc–dc power converters depends strongly on the voltage and power levels as well as application requirements. Weight reduction is particularly important for aerospace applications [8]–[13], [23], [24], [27]–[30] and, therefore, has driven research on high-voltage dc–dc power converters having high specific power for space and aircraft applications. For megawatt, hundreds of kilovolt power converters, specific power can reach as high as $10 \text{ kW} \cdot \text{kg}^{-1}$ [12].

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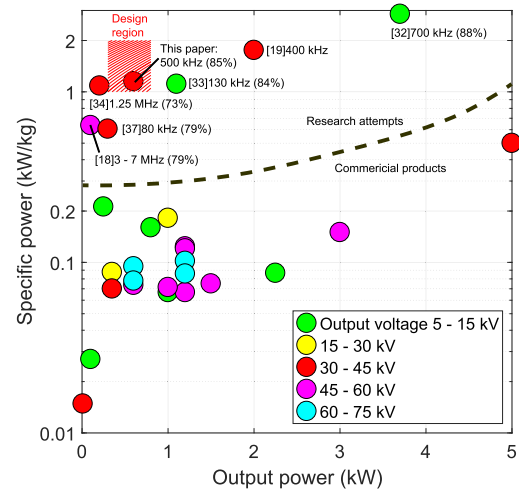


Fig. 1. Specific power of commercial high-voltage dc–dc converters and academic designs at low-to-moderate power levels. The reported switching frequencies and efficiencies are listed next to the citations. The design region marks the requirements of interest in this article.

For tens of kilovolts and tens of kilowatts, $1 \text{ kW} \cdot \text{kg}^{-1}$ has been achieved in research papers [12], [31]. However, there has been less research in increasing the specific power of such converters in the subkilowatt and medium-to-high tens of kilovolt range.

In Fig. 1, a review of commercial products and academic designs in this power and voltage range reveals that the specific power of commercial converters is typically around $0.1 \text{ kW} \cdot \text{kg}^{-1}$. In these converters, the switching frequency typically lies around 100 kHz or lower, resulting in bulky magnetics and capacitors. There have been research attempts (see [18], [19], and [32]–[35]) to increase the switching frequency and thus the specific power, but these have largely focused on cases of relatively low output voltage ($\leq 20 \text{ kV}$) [32], [33], [35] and/or out of the power range of interest ($\geq 2 \text{ kW}$ [19], [32] or $\leq 200 \text{ W}$ [18], [34]). See [36] for the detailed list of products and academic papers in this comparison.¹

Aerospace applications requiring low-power high-voltage conversion are emerging, in which specific power is a major consideration. For example, electroaerodynamic (EAD) aircraft propulsion, which uses a set of “solid-state” electrodes to generate ions in air and accelerate the ions to produce a thrust force for propelling aircraft, requires lightweight high-voltage power [2], [3], [38], [39]. Most EAD systems use a corona discharge

¹Precise weights of academic designs are not documented in [19], [32], [33], and [37]; thus, the authors estimated these numbers from info in the papers.

to produce ions, whereby a high-potential gradient near a set electrodes with high radius of curvature (e.g., needlepoint electrodes) results in a self-sustaining production of ions, although systems with other ion sources have been demonstrated [40], [41]. In both cases, voltage levels of tens of kilovolts are required to produce useful levels of thrust [2]–[4]. Since EAD propulsion has no moving parts and does not require propellers or turbines to produce thrust, it has the potential to make aircraft quieter, safer, and more robust. In particular, EAD propulsion has shown promise in unmanned aerial vehicle (UAV) applications [2]–[4]; however, no EAD UAV was ever flown prior to the work by Xu *et al.* [42], in large part due to the weight of the required power electronics. For the EAD propulsion system of interest, the desired voltage and power are identified to be medium-to-high tens of kilovolts and hundreds of watts [2]–[4]. The specific power of the converter needs to be $\geq 1 \text{ kW} \cdot \text{kg}^{-1}$ and still higher is preferable as technology allows.

This article explores the design of a lightweight high-voltage power converter suitable for EAD propulsion systems. A high-voltage converter typically consists of three stages: an inverter stage, an isolation/transformation stage, and an ac–dc rectifier stage. This article first compares different approaches and topologies to realize each stage in terms of the resulting weight. Furthermore, the overall weight of a converter based on the best identified approach is optimized by sweeping through combinations of voltage gains for the different stages with considerations of device limitations and losses. The optimized converter design is selected, which consists of three stages: a series–parallel resonant inverter, a high-voltage transformer, and a full-wave Cockcroft–Walton multiplier. The optimized stage contributes nominal voltage gains of about $2.5\times$, $\sim 15\times$, and $\sim 5.6\times$, respectively, for a nominal voltage gain of 210 and a maximum voltage gain of 250. A prototype converter is constructed and tested converting a battery voltage of -225 V to a 40-kV output voltage and providing up to 565 W . At the peak output voltage and power, it switches at $\sim 500 \text{ kHz}$, achieves an efficiency of 85% and a specific power of $1.15 \text{ kW} \cdot \text{kg}^{-1}$. The prototype was used to enable the first flight of an EAD aeroplane [42]. This article also explores approaches that may be exploited to achieve even higher specific power of such high-voltage low-power converters. Section II explores the design goals at a high level, analyzing tradeoffs of different approaches and topologies to realize each stage. Narrowing down to the specific design target of the EAD application, Section III selects and optimizes the topology of the converter, taking into consideration the practical constraints of available devices. Section IV describes practical issues in building such a lightweight high-voltage dc–dc converter prototype and presents experimental results of the prototype converter. Section V presents ways to further increase the specific power. Finally, Section VI concludes this article.

II. TOPOLOGY COMPARISON

High-voltage step-up dc–dc converters achieve a large step-up voltage ratio using a combination of: 1) resonant and/or multi-level inverters; 2) large-step-up-ratio isolation stage; 3) voltage multiplier rectification stage; and 4) parallel-input series-output

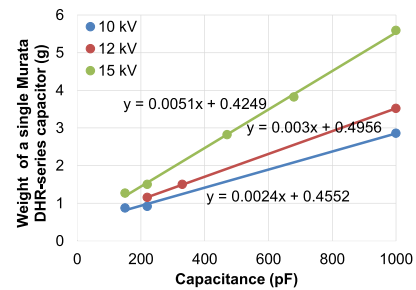


Fig. 2. Capacitance and voltage dependence of Murata DHR series high-voltage capacitors (the dots are the measured weights and the lines are linear fits).

structures [43]. This section compares the weight of different topologies/approaches to realize each stage. Section II-A compares the weights of four voltage multiplier topologies considering the characteristics of available high-voltage capacitors and diodes. Section II-B compares the weight of conventional cored transformers, resonant transformers, and piezoelectric transformers. Section II-C briefly analyzes the tradeoffs of each resonant inverter topology.

A. Voltage Multiplier

The achievable weights of voltage multipliers depend substantially on available high-voltage capacitors and diodes.

1) *High-Voltage Capacitor Selection:* Mica, film, and ceramic capacitors are commonly used in high-voltage converters. For high-voltage low-power applications, the rated voltage, the capacitance, and the weight of the capacitors are three main considerations. Their equivalent series resistance and current-carrying capability are less important because of the relatively small output current.

Capacitor weight varies widely across rated voltage, capacitance, materials, package, and manufacturer. When the rated voltage is below $\sim 5 \text{ kV}$, there are surface mount technology (SMT) options in all three material classes, among which ceramic capacitors with similar capacitance yield slightly lower weight. When the rated voltage is above 8 kV , there are limited options. Ceramic capacitors with leads offer medium capacitance (up to 1 nF) and relatively high rated voltage (up to 15 kV) and are much lighter options compared with ceramic screw-mount, Mica, and film capacitors.

To realize a capacitor blocking above 8 kV , the two lightest options are: 1) using ceramic SMT ones rated below 5 kV in series and parallel, and 2) using a single ceramic through-hole one.

Among the investigated parts, Murata DHR series ceramic through-hole capacitors have the least weight in the $6.3\text{--}15 \text{ kV}$ range. At a given rated voltage, the weight of the capacitor is shown to be proportional to its capacitance; thus, a linear relationship is extrapolated for other capacitances at this voltage rating.² See Fig. 2 for examples in the range of $10\text{--}15 \text{ kV}$ and [36] for more details.

²In the same family from the same manufacturer. Note that the linear fits vary significantly for other manufacturers, series, and materials.

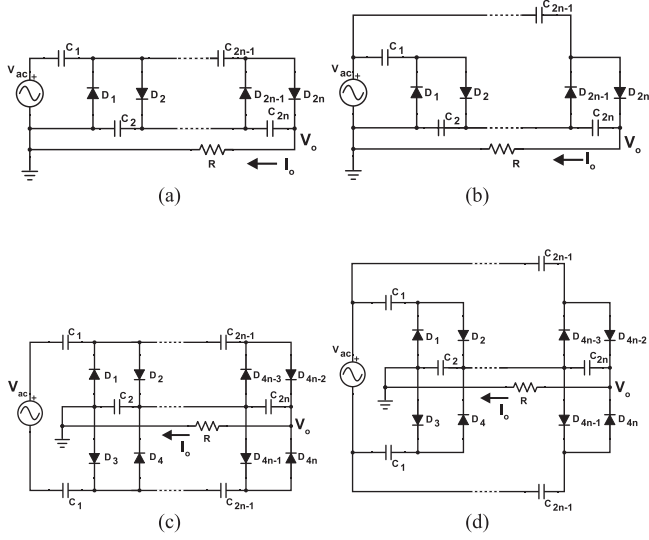


Fig. 3. Four basic voltage multiplier topologies (showing single polarity). (a) Half-wave Cockcroft-Walton. (b) Half-wave Dickson. (c) Full-wave Cockcroft-Walton. (d) Full-wave Dickson.

2) *High-Voltage Diode Selection*: High-voltage diodes are a key limitation in building a high-frequency high-voltage converter at tens of kilovolts and hundreds of watts. Traditional silicon high-voltage diodes (≥ 8 kV) have longer recovery times compared to their low-voltage counterparts and, thus, are typically used at frequencies below 100 kHz. Commercially available and affordable high-voltage SiC Schottky diodes are mostly rated under 3.3 kV. They exhibit little recovery time³ but large parasitic capacitance since they are mainly designed for high-current applications and have a large die size.

Cree's C4D02120A, GeneSiC's GAP3SLT33-214, and VMI's X150FF3 are considered in later studies in Section III because of their small recovery time and relatively low parasitic capacitance. See other device types considered in [36] and [44]. There have also been emerging SiC p-i-n diodes 8- and 15-kV ones from GeneSiC that could be potential options if cost were not a consideration.

The nominal design frequency is chosen to be 500 kHz considering steep increases in loss that are observed for many diodes (including the one selected) at higher frequencies.

3) *Weight Study of Various Voltage Multiplier Topologies*: The typical topologies of the voltage multipliers used in high-voltage converters are shown in Fig. 3. All flying capacitors are odd-indexed and all output capacitors are even-indexed.

We approximate the weight of the voltage multiplier as the sum of the weights of its capacitors. The weights of multipliers in each topology achieving a 40-kV 700-W output at different input voltages (thus different gains) are compared in Fig. 4(a).

At the power and voltage levels in this application, and with the device considerations described above, Cockcroft-Walton topologies yield similar weight with Dickson topologies but have

³There is a nonzero "switching time" listed in the datasheet of some of these diodes (for example, the 3.3-kV SiC Schottky diode GAP3SLT33-214).

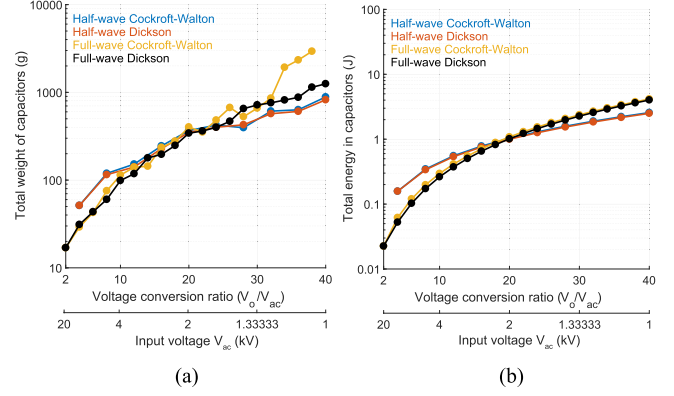


Fig. 4. (a) Total weight of the capacitors and (b) total energy stored in the capacitors of the Cockcroft-Walton and Dickson voltage multipliers as a function of conversion ratio (20-kV 350-W output).

easier implementation.⁴ When the required voltage gain is lower than 10, full-wave topologies yield lower total weight (owing to interleaving reducing output capacitance), and so are strongly preferred; when the required voltage gain is higher than 25, half-wave topologies yield lower total weight (because they provide the same voltage gain with a halved number of stages, and a further halved flying capacitor count, compared with their full-wave counterparts). This trend coincides with the total energy storied in the capacitors [see Fig. 4(b)]. The variations between the weight and the energy can be explained by the discrete nature of the capacitors' energy density.

The process of the weight comparison is illustrated as follows: we first consider bipolar (or interleaved) voltage multipliers, with each polarity containing n stages to process half of the desired power and voltage. Then, we pick a topology and calculate the weight of each capacitor. To do so, we identify the voltage across each capacitor and calculate their capacitances following two specifications below. An assumption is made to simplify the calculation: in a given topology, all flying capacitors (odd-indexed) are the same and all output capacitors (even-indexed) are the same. For example, in a half-wave Cockcroft-Walton voltage multiplier, $C_1 = C_3 = \dots = C_{2n-1} \triangleq C_{\text{odd}}$ and $C_2 = C_4 = \dots = C_{2n} \triangleq C_{\text{even}}$.

- 1) The even-indexed capacitors in half-wave topologies are sized such that the voltage ripple is less than 100 V (see [36] for the voltage ripple as a function of even-indexed capacitances). Ideally, for full-wave topologies, the ripples of the two half-wave would cancel each other, and no output capacitors are needed. However, to account for nonidealities, we size these even-indexed capacitors such that the RC time constant of the effective total output capacitance and the load is 20 times the switching period.
- 2) The odd-indexed capacitors in all topologies are then sized such that the voltage droop due to the capacitor charge loss [45] is less than 2.5% of 20 kV (5% for the full converter outputting 40 kV). See [36] for the voltage droops as functions of capacitances.

⁴In Cockcroft-Walton topologies, capacitors and diodes block the same voltage other than the flying capacitor in the first stage, whereas in Dickson topologies, the blocking voltage increases in higher stages.

TABLE I
WEIGHT COMPARISON OF DIFFERENT ISOLATION STAGES

Type	Rated power	Rated voltage	Total weight	Specific power
Ferrite core transformer	1 kW	10 kV	250 - 320g †	3.1 - 4kW kg ⁻¹
Tesla coil	1 kW	10 kV	135 - 152g ‡	6.6 - 7.4kW kg ⁻¹
Piezoelectric transformer	Use STEMiNC SMMTF53P2S40 2W 1.2 kV piece, 9S60P [50]		1080	0.93
	Use Face Transonor 1.5 kV piece, 7S17P [51]		1095 † ‡	0.92

†bounds correspond to different insulation materials. Lower bound is Nylon 6/6 (glass) and upper bound is PTFE (glass).

‡Bounds correspond to whether to include a supportive structure for the secondary winding made with a 0.1-mm-thick Nylon 6/6 tube.

††The unit weight is estimated with the same density as [50] and a size of 120 × 8 × 5.7 mm. Assume 7 in series 17 in parallel.

Then, for each capacitor, we calculate its weight following the linear relationships with two assumptions: 1) the rated voltage of the capacitor is twice its required blocking voltage and 2) we allow up to four discrete capacitors to connect in series and 20 in parallel to achieve the desired blocking voltage and capacitance. Finally, we sum up the capacitor weight and double it (due to the bipolar design) to obtain the “total weight” of the voltage multiplier. The total energy stored in capacitors is calculated in the same fashion.

B. Isolation Stage

Three isolation methods are compared in terms of weight: full-core transformers, air-core resonant transformers, and piezoelectric transformers.

Full-core transformers dominate the isolation stage design in traditional high-voltage converters. A study in [12] for high-power low-frequency transformers (0.1–100 MW, tens of kilohertz, no voltage insulation requirements were considered) shows that the theoretical specific power of transformers roughly scales with its frequency f and power P as $f^{0.75} P^{0.25}$, indicating that specific power improves with higher power. Air-core resonant transformers (e.g., Tesla coil) eliminate the use of a heavy core and could potentially be light. However, it is a highly tuned structure, sensitive to parasitic inductance and capacitance, and may be relatively large in size [46], [47]. Piezoelectric transformers can achieve a power density of 40 W·cm⁻³ [48], especially suitable for high-voltage low-power (e.g., tens of watts) applications. They have been used in space applications at below 20 kV and 200 W [49]. However, when scaling up in power, one needs to connect existing piezoelectric transformers in series and in parallel, and the advantage in power density is less obvious.

To illustrate these tradeoffs, isolation stages were designed, and the resulting weights were compared, as shown in Table I. Full-core and air-core transformers were both designed to step up a 500-V amplitude 500-kHz ac to a 10-kV amplitude ac, delivering 1 kW. See [36] for detailed designs.

The full-core transformer design yields a specific power of 3.1–4 kW · kg⁻¹. The choice of the insulation material around the secondary wire and the core affects the total weight and the winding’s parasitic capacitance. Here, we give a lower bound of

250 g where glass-filled Nylon 6/6 is used and an upper bound of 320 g where glass-filled polytetrafluoroethylene (PTFE) is used.

The air-core resonant transformer, designed following [46] and [47], yields 135 g and can reach 152 g and higher with additional supportive structure. The air-core resonant transformer shows doubled specific power compared with the full-core transformer, but is less efficient. This would require heavier energy source (e.g., battery) in the intended EAD application. In addition, it is necessarily large in size, is less robust in construction, and requires narrowband operation, which makes this option less appealing.

Piezoelectric transformers are designed to provide the same output voltage, 10 kV in amplitude, and output power, 1 kW, by connecting existing piezoelectric transformers from Transonor [51] and STEMiNC [50] in series and in parallel. We leave the input voltage parameter uncontrolled. As an example, we use SMMTF53P2S40 (rated for 2-W 1.2-kV output): the weight and the dimension of a single off-the-shelf part are 2 g and 35.8 × 8.8 × 3.8 mm; thus, its density is 1.67 g·cm⁻³. To achieve 10 kV and 1 kW, we need nine modules in series and 60 in parallel, yielding 1080 g and 0.93 kW · kg⁻¹.

Considering both feasibility and achievable specific power, full-core transformers are compact, robust, and more flexible in use than air-core resonant transformers or piezoelectric transformers at the voltage and power level, so are preferred.

C. Resonant Topology Selection

High-voltage transformers have a large number of secondary turns, resulting in large parasitic capacitances. To usefully incorporate this capacitance into circuit operation, a parallel resonant inverter or a series–parallel resonant inverter is suitable. A full-bridge series–parallel resonant inverter is chosen because: 1) it provides a factor of 2 in the voltage gain with negligible weight gain; 2) it shows high efficiency in both light load and heavy load; and 3) it has a series capacitor to block dc voltage and, thus, prevent transformer saturation [52], [53].

III. DESIGN OPTIMIZATION

Based on the weight study and engineering considerations such as risks and practicality, a core transformer and a series–parallel resonant inverter are chosen. Different distributions of voltage gains among the three stages can result in different overall weights. Section III-A explains that in our design, the available diodes set the transformer secondary voltage, which decouple the voltage multiplier stage from the overall optimization. Section III-B comprehensively designs the inverter and transformer stages to minimize the total weight of the passives while maintaining a good efficiency.

A. Finalizing the Voltage Multiplier

In practice, the transformer parasitic capacitance and the junction capacitance of the diodes are reflected to the transformer primary with a multiple of the square of the turns ratio. Too

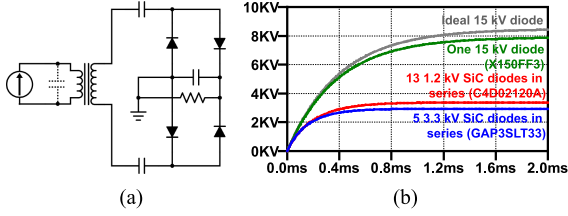


Fig. 5. (a) Simulation schematics and (b) the output voltage of three single-stage full-wave Cockcroft–Walton multipliers using three different diodes. In the schematic, the dashed-line capacitor represents the parasitic capacitance reflected to the transformer primary.

much these capacitance can result in large circulating current in the resonant tank and, thus, lower efficiency.

Using Cree’s C4D02120A, GeneSiC’s GAP3SLT33-214, and VMI’s X150FF3, respectively, a single-stage full-wave Cockcroft–Walton voltage multiplier is simulated in LTspice [see Fig. 5(a)]. Each voltage multiplier is driven by a 500-kHz 0.5-A current source and uses the same transformer. The diode in the voltage multiplier consists of either one 15-kV X150FF3 or 13 1.2-kV C4D02120A or five 3.3-kV GAP3SLT33, yielding a diode blocking voltage of ~ 15 kV. As a reference, a voltage multiplier with ideal 15-kV diodes that have no forward voltage drop or parasitic capacitance is also simulated.

The simulation results in Fig. 5 show that the voltage droop, defined as the difference between the output voltage when using ideal diodes and those when using investigated diodes, is the lowest with X150FF3 diodes. Using multiple SiC diodes theoretically reduces the overall parasitic capacitance, but since their junction capacitance is much larger than that of X150FF3, the simulation still shows unacceptably large droop. In addition, series-connecting multiple SiC diodes can potentially require balance circuits and add more complexity. Therefore, in this design, the X150FF3 is selected as the most effective available diode. To simplify the design and because of second-order effects that can occur [44], we do not consider connecting multiple of X150FF3 in series as an equivalent higher voltage diode, which sets the output of the transformer to be less than ~ 7.5 kV (considering a 50% voltage derating).

Referring back to the voltage multiplier weight study in Fig. 4(a), a three-stage bipolar (six-stage total) full-wave Cockcroft–Walton voltage multiplier can convert 7.5 to 40 kV and yield a light weight. Compared to a three-stage full-wave Dickson, the Cockcroft–Walton voltage multiplier has a simpler implementation because most flying capacitors have the same capacitances and block the same voltages; thus, it is chosen.

B. Weight Study of the Inverter and the Transformer

The series–parallel inverter and the transformer stages (see Fig. 6) are required to convert $V_{DC} = 200$ V dc to $V_{Sec} = 7.5$ kV ac at 500 kHz. The transformer is chosen to be center-tapped so that it handles a bipolar output voltage rather than a unipolar output voltage, which reduces the isolation voltage requirement by half. To account for the losses in the voltage multiplier, we design at $P_{Sec} = 750$ W rather than 700 W. To simplify the weight analysis, we use the fundamental harmonic approximation [52]

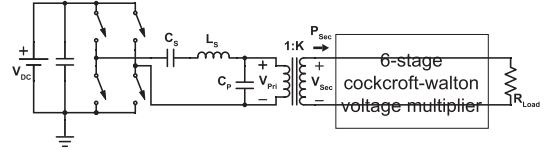


Fig. 6. Inverter and the transformer stages. The resonant tank input is a square wave of amplitude V_{DC} containing no dc component.

method to analyze the inverter operation and assume V_{Pri} to be sinusoidal.

The weight of the two stages is dominated by magnetics (the inductor and the transformer); thus, the design space is explored to minimize the summed weight while maintaining good efficiency.⁵ The optimization process contains five steps.

- 1) Select a set of operating variables: the resonant tank quality factor Q , the tank natural frequency f_0 , the series–parallel resonant capacitance ratio $A = C_P/C_S$, and the transformer voltage step-up ratio K (i.e., turns ratio). Then, the inductance and capacitances of the resonant tank (L_S, C_S, C_P), the tank voltage gain $G = V_{Pri}/V_{DC}$, and the tank current $I_{L_{max}}$ can be calculated (see [36] for details). We keep the sets that yield a transformer output voltage $V_{Sec} = V_{DC} \times G \times K \in [7.5 \text{ kV}, 9 \text{ kV}]$.
- 2) Design the lightest transformer that satisfies an input voltage of $V_{DC} \times G$, turns ratio K , output power $P_{Sec} = 750$ W, and a set of constraints on efficiency, temperature, and packing factor. See [36] for detailed methods.
- 3) Design the lightest inductor with inductance L_S , maximum current $I_{L_{max}}$, and a set of constraints on efficiency, temperature, and packing factor. See [36] for details.
- 4) Sweep all sets of operating variables and find the lightest summed weight of the inductor and the transformer. The design space we explored is $0.2 \leq Q \leq 10$, $0.1 \leq A \leq 10$, $450 \text{ kHz} \leq f_0 \leq 500 \text{ kHz}$, and $5 \leq K \leq 30$.
- 5) Refine the lightest designs by checking the parasitics of the transformer: 1) if the transformer leakage inductance is not negligible compared with L_S , then redesign the inductor; and 2) if the transformer parasitic capacitance is larger than C_P , then section its secondary winding.

To simplify the study of this prototype, we consider: 1) center-leg winding on E-type cores for both the transformer and the inductor to ensure good coupling;⁶ 2) only off-the-shelf core sizes. Customized cores may yield more optimal designs and will be discussed in Section V; and 3) the secondary wire fixed as Teledyne Reynolds AWG28 18 kV FEP wire.⁷

C. Optimal Weight and Voltage Gain Distribution

In Fig. 7, we plot the lightest overall weights of magnetic components and their corresponding inductor and transformer weights against the transformer primary voltages V_{Pri} in the range of 200–1800 V, which is a function of Q , A , f_0 , and K .

⁵The transformer is to be $\geq 95\%$ efficient and the inductor is to be $\geq 98\%$.

⁶This makes the transformer leakage inductance small compared with L_S .

⁷P/N 178-5790; this is the smallest gauge high-voltage wire (≥ 15 kV) we found in a time limited search. Its voltage rating and gauge size may be conservative but provide good design margin.

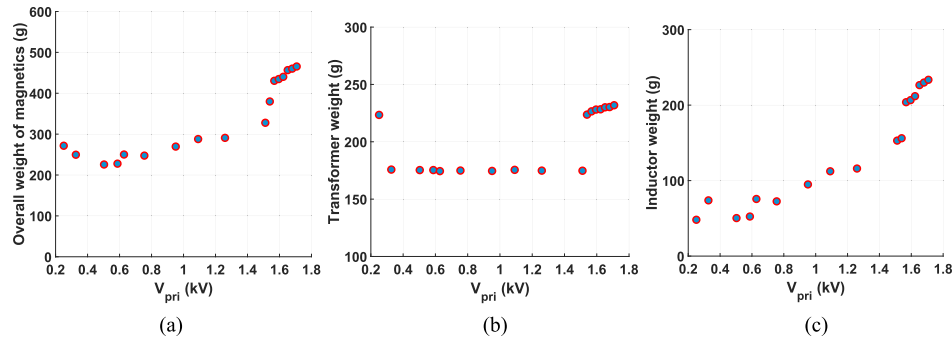


Fig. 7. Lightest overall weights of (a) both the inductor and the transformer, (b) the corresponding inductor weights, and (c) the corresponding transformer weights against the transformer primary voltages V_{pri} . The discreteness is due to our assumptions to use discrete off-the-shelf core shapes, wires, and discrete operating variables Q , A , f_0 , and K . Customized cores and wires can yield to a lighter overall weight, as will be discussed in Section V-B.

The lightest overall weight shows a clearer trend with V_{pri} than any of Q , A , f_0 , and K and, thus, is plotted.

Fig. 7(a) suggests that the overall weight of magnetic components first decreases and then increases with V_{pri} . It reaches the lowest when V_{pri} is around 400–600 V. For this range of V_{pri} , the resonant tank provides a gain of $G \sim 2 - 3\times$ and the transformer provides a gain of $K \sim 18 - 12\times$. A valley point is reached because of how the core sizes required for the inductor and the transformer trade with the voltage gains provided.

1) The transformer weight [see Fig. 7(b)] first decreases and then increases with V_{pri} . All designs are temperature limited.

a) When $V_{pri} \leq 400$ V, a larger turns ratio of the transformer requires a larger amount of windings and a bigger core for the windings, yielding a heavier weight.

b) When V_{pri} is in 400 V–1.5 kV, the volt-second on the core keeps increasing, but the transformer weight stays relatively flat because: 1) to keep the maximum flux density B_m low, we need a higher primary number of turns N_p , but higher V_{pri} results in lower turns ratios, yielding unchanged (or even reduced) number of secondary turns; thus, the copper volume and weight stay relatively flat; and 2) the core volume and weight also remain unchanged because no bigger cross-sectional area is needed to balance the increasing volt-second, and similar cores can be used to fit the relatively unchanged winding volume.

c) When $V_{pri} \geq 1.5$ kV, the volt-second keeps increasing; the original core height is not enough to fit the increasing amount of primary number of turns N_p ; thus, a larger core with bigger window height is needed, yielding a heavier weight.

2) The inductor weight [see Fig. 7(c)] increases with V_{pri} , especially when $V_{pri} \geq 0.8$ kV (corresponding to the tank voltage gain $G \geq 5$ and tank quality factor $Q \geq 3$). All designs are also temperature limited. At higher tank gain G , to keep the maximum flux density low, more turns and bigger core cross-sectional area are required, yielding heavier core and copper weights.

For the final design, we pick V_{pri} to be at ~ 500 V, the resonant tank Q is ~ 2 , the tank voltage gain $G \sim 2.5$ and the transformer turns ratio is $K = 15$. The final converter topology is shown in

Fig. 8. It comprises a series-parallel resonant inverter, a 1:15:15 ferrite-cored high-voltage transformer, and a six-stage bipolar full-wave Cockcroft–Walton multiplier.

D. Sectioning the Transformer Secondary Winding

When designing the transformer, an important criterion is to use its parasitic capacitance as part of the parallel resonant capacitance C_p . A multisection secondary is a common solution to reduce the self-capacitance of the winding. A rule of thumb is that with n sections, the self-capacitance can be reduced to $1/n$ [54]. The winding-to-core capacitance is considered to be not affected by any sectioning. See [36] for detailed equations.

In this study, we consider one multisection secondary, and the section number is swept between 1 and 4. In the final design, the transformer secondary is sectioned into two sections. No higher number of sections is needed.

IV. EXPERIMENTAL RESULTS

A prototype converter with closed-loop voltage feedback control based on the optimized design in Fig. 8 was built, as shown in Fig. 9. All the sensing, control, and driver circuits are integrated on the printed circuit board, and it needs no additional components other than a logic power supplied by a 3.7-V LiPO battery for the EAD flight demonstrations. There have also been special considerations in construction of the voltage multiplier (include heatsink design, FEA thermal, and field analysis). See [36] for details. We also explicitly used air isolation as a strategy, which improves the specific power at the expense of the power density.

Fig. 10 shows the waveforms of the converter running at 500 kHz and converting 177 V to 38 kV at ~ 480 W. The converter achieves an efficiency of 83% at 480 W and 38 kV and 82% at 300 W and 40.4 kV. We were not able to push to above 550 W output before the MOSFETs GS66504B overheated. A previous version of the inverter using EPC 2025 MOSFETs was tested at 565-W 39-kV output, achieving 85% efficiency. EPC 2025 went obsolete; therefore, we switched to GS66504B. The largest component of loss is estimated to result from the high-voltage diodes, followed by the transformer, with MOSFETs and inductor losses being smaller. Moreover, the diode loss was the dominant limitation preventing increases in frequency and

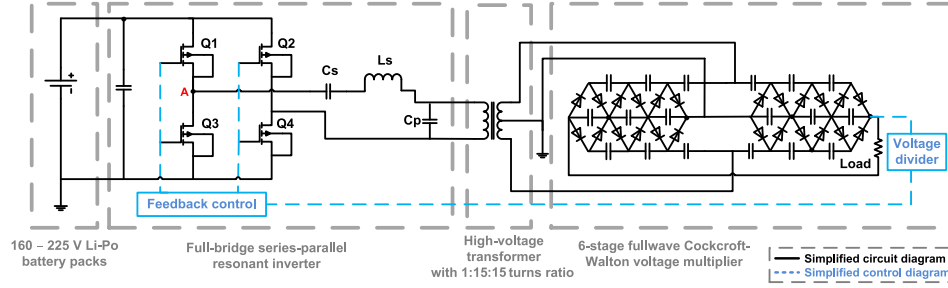


Fig. 8. Final design of the developed high-voltage dc-dc converter. Node A marked in red corresponds to the switching node waveform in Fig. 10.

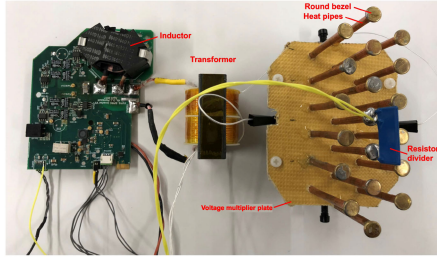


Fig. 9. 200-V to 40-kV high-voltage dc-dc converter prototype.

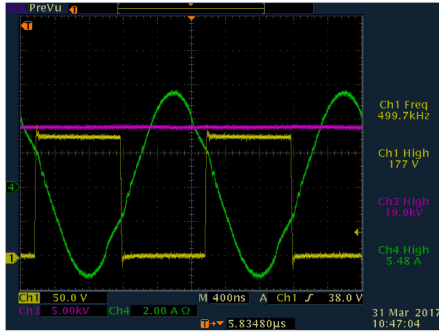


Fig. 10. Experimental waveforms of the developed high-voltage dc-dc converter prototype outputting ~ 480 W and 38 kV (Ch1 (yellow): half-bridge switching node voltage (Node A marked in red in Fig. 8); Ch3 (pink): half of the output voltage; and Ch4 (green): resonant tank current).

specific power in this design, resulting in the diode investigation in [44].

The prototype was used to drive electrode thrusters of an EAD aeroplane and demonstrated the first flight of such an airplane [42]. The finalized components in the converter and the weight breakdown are listed in Table II. The prototype achieves a specific power of $1.15 \text{ kW} \cdot \text{kg}^{-1}$ ($565 \text{ W}/491 \text{ g}$) and a box power density of $0.56 \text{ W} \cdot \text{cm}^{-3}$ ($565 \text{ W}/1007 \text{ cm}^3$). In this design, the specific power is a more important metric, given that the main goal is to reduce the weight rather than the volume of the converter.

V. PATHS TO FURTHER IMPROVE THE SPECIFIC POWER

In exploring future designs of EADs, we identify two new requirements for the high-voltage converter:

- 1) higher output voltage at 40–60 kV or even higher with:
 - 1) similar input voltage, 2) similar output power of up to 600 W, and 3) similar (85%) or higher efficiency;

- 2) higher specific power at or above $1.5 \text{ kW} \cdot \text{kg}^{-1}$.

We present two ways to further improve the specific power of a converter that can convert 200 V to 60 kV at 600 W.

- 1) In Section V-A, we incorporate insights learned about high-voltage diodes in [44] in the design. This allows us to remove cooling fixtures and their associated weights and increase the switching frequency to $\geq 1 \text{ MHz}$, which further reduces the size and weight of passives.
- 2) In Section V-B, we explore more flexible high-voltage transformer designs that may yield lower weight, including customized core size and core shape, different winding patterns, and various high-voltage wires.

A. Improved Voltage Multiplier Design

One of the bottlenecks to achieving high frequency at high output voltage while preserving high efficiency is the lack of low-loss high-frequency high-voltage diodes [55]. In [44], we tested 37 off-the-shelf diodes in a full-bridge rectifier switching at both 600 kHz and 1 MHz, and each diode blocking 50% of their rated voltage and carrying an average current 5–10% of their rated dc current (we call these “single diode tests”). We identified 17 diodes with a maximum temperature rise lower than 100°C at both frequencies. These diodes are promising candidates for the improved voltage multiplier.

We want to narrow down the selections to the diodes that are both lighter weight and less lossy. We use the temperature rises of each diode at 600 kHz and 1 MHz in the single diode tests as their loss metrics. We define a weight metric for each diode as the unit weight of the diode multiplied by the number of diodes needed in series to block a dc voltage V (with a 50% derating on the rated voltage) and then the number of diodes needed in parallel to carry a dc current I (with a 10% derating on the rated average forward current). We pick $V = 60 \text{ kV}$ and $I = 20 \text{ mA}$ to represent high-voltage low-current applications.

Fig. 11(a) and (b) plots the defined weight metric against the defined loss metric at 600 kHz and 1 MHz, respectively. Each data point is labeled with the diode index. Diode 18 (X150FF3) is used in the first-generation high-voltage dc-dc converter. Details of these diodes are listed in Table III.

We prefer diodes with a temperature rise $\leq 40^\circ \text{C}$ and a weight metric $\leq 20 \text{ g}$ at both frequencies (highlighted in green). We pick 40°C because 1) this temperature rise is likely a conservative estimate of the actual temperature rise of the diodes when connected in series and/or used in a multistage voltage multiplier;

TABLE II
SPECIFICATIONS AND COMPONENT WEIGHT BREAKDOWN OF THE PROTOTYPE HIGH-VOLTAGE POWER CONVERTER

Stage	Component	Manufacturer and Part Number	Value/Description	Weight	Weight %
Inverter	MOSFETs	GaN system GS66504B	-	56 g	11.4%
	C_s	TDK C3216C0G Series	19.6 nF		
	C_p	Parasitic capacitance	~5 nF		
	Inductor L_s †	RM14I core in TDK N49; MWS wire AWG14(150/36)	11 turns, air gap 0.87 mm, 33.2 μ H	80 g	16.3%
Transformer	Core ‡	ETD49/25/16, Ferroxcube 3F35	-	124 g	In total, 170 g, 34.6%
	Primary†	MWS AWG 16 (350/42)	10 turns, 1 layer; Total length 0.52 m; Unit weight 18.45 g m ⁻¹	9.6 g	
	Secondary†	Teledyne Reynolds AWG28 18kV FEP wire (P/N 178-5790)	150 turns, 2 section. In each section, 5 layer 15 turns per layer; Total length 12.5 m; Unit weight 2.68 g m ⁻¹	34 g	
	Bobbin	ABS 3D printed	-	2.4 g	
Voltage Multiplier	C_{odd}	Murata DHR4E4B102K2BB	1 nF. Unit weight 3.3 g. Qty. 12.	40 g	In total, 115.3 g, 23.5%
	C_{even}	Murata DHR4E4B681K2BB	0.68 nF. Unit weight 2.3 g. Qty. 6.	14 g	
	Diodes	VMI X150FF3	Unit weight 0.51 g. Qty. 48.	25 g	
	Bezel joints	Metalliferous BR8818	Unit weight 0.8 g with solders. Qty. 19.	31 g	
	Divider	OHMITE SM204RD-0009	100M/100k 1000:1 resistor, 1%	1.7 g	
	Connectors	Pomona Electronics 5936-0 and 5935-0	Unit weight 0.6 g. Qty. 6.	3.6 g	
	Output wire	Teledyne Reynolds AWG22 30kV Silicone coated FEP wire (P/N 178-8781)	-	-	
	Heat sink	Wakefield-Vette 121686_R EV1. 7mm long 4mm OD.	Unit weight 3.6 g. Qty. 19.	69 g	14%
Total weight				491 g	100%

The total weight here does not include that of the supportive structures and the peripheral circuits, which are necessary for the flight.

†The same core in TDK N49 would yield to slightly lower loss but not available.

‡The number of primary turns is reduced to 10 and the secondary correspondingly to 150 (1:15), compared with the designed 13 and 196 (1:15) due to imperfection in hand-winding and added insulation thickness in the practical construction.

TABLE III
SPECIFICATIONS OF CANDIDATE DIODES IN THE SECOND-GENERATION VOLTAGE MULTIPLIER DESIGN

Diode index	Manufacturer	Part Number	Rated voltage (V)	Rated current (mA)	Type	Unit weight (g)	Length (mm)	Width (mm)	Temperature rise of one diode (°C)		Loss of one diode at 1 MHz (W)	Test current at 1M (mA)	Forward drop (V)	Capacitance at 0V (pF)
1	Wolfspeed	CSD01060E	600	1000	SiC Schottky	0.315	6.73	10.00	9.40	8.40	0.07			
2	Infinion	C3D1P7060Q	600	1700		0.033	3.30	3.30	11.40	12.30	0.08	200.00	2.4	82.5
3	GeneSiC	GB01SLT06	650	1000		0.088	5.59	3.94	10.3	8	0.06			
4	Infinion	IDL02G65C5	650	2000		0.184	8.10	8.10	10.5	10.5	0.09			
5	GeneSiC	GB01SLT12	1200	1000		0.088	5.60	3.95	9.8	16	0.12	200.00	2.4	71.0
6	Infinion	IDM02G120C5	1200	2000		0.315	6.73	10.00	10	16.7	0.13			
7	ST	STPSC5H12B	1200	5000		0.390	6.73	10.00	13	22	0.22			
8	GeneSiC	GAP3SLT33-214	3300	300		0.065	5.59	3.94	16.5	40	0.39	30.00	5.2	38.0
9	Bourns Inc.	CD214A-F1400	400	1000	Si	0.068	5.59	2.92	10.5	19.7	0.18			
10	Taiwan	ESH1GM	400	1000		0.006	2.70	1.35	13.6	16.5	0.10			
11	Taiwan	UF1GLW	400	1000		0.014	3.80	1.90	14.9	17.5	0.19	200.00	1.3	25.0
12	VISHAY	BYV26B	400	1000		0.161	4.00	3.60	21	31	0.31			
13	ROHM	RFU02VSM6S	600	200		0.006	2.50	1.40	6.8	10.1	0.06			
14	VISHAY	BYV26C	600	1000		0.155	4.00	3.60	20	44.5	0.56			
15	ROHM	RFU02VSM8S	800	200		0.007	2.50	1.40	23.6	77	0.32			
16	Dean	SP5LFG	5000	270		0.102	8.55	2.79	47.2	93.8	0.53			
17	Dean	HVEF8P	8000	30		0.082	10.20	2.50	21.9	28.5	0.19	6.00	20.0	0.3
18	VMI	X150FF3	15000	50		0.510	9.14	4.32	75					

The ones highlighted in red are the most promising. Temperature data, loss data, and the manufacturer abbreviation are all the same as in [44].

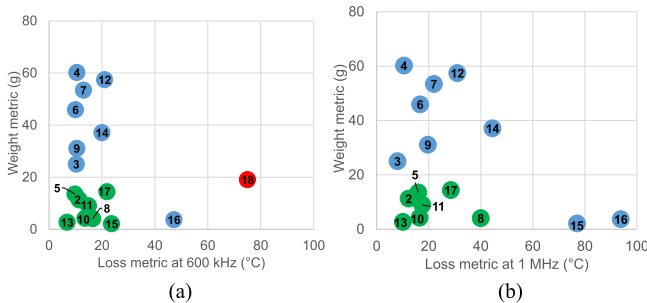


Fig. 11. Defined weight metric of each diode plotted against the defined loss metric at (a) 600 kHz and (b) 1 MHz for realizing an effective rectifier device for use at 60 kV and 20 mA. The ones in green are promising. Diode No. 18 is used in the first-generation converter.

and 2) when it is above 40 °C, a heat sink is likely required. We pick 20 g because there seems a clear separation between 20 g and the next lightest.

In addition, in [44], we also tested these diodes in the same full-bridge rectifier, but with each rectifier leg consisting of several diodes connected in series. Higher temperature rises are observed due to the imbalance in voltage distribution among the series-connected diodes. Diodes 10 and 13 exhibit much severer temperature and voltage imbalance when connected in series and, thus, are removed.

We narrow down to five diodes (no. 2, 5, 8, 11, and 17) as candidates. With the identified diodes and following similar design methods in Section II, we built weight models for various voltage multipliers, which were feed into the overall weight model of the dc–dc converter. See [36] for details.

B. Updates on High-Voltage Transformers

We identify two factors affecting the achievable weight and specific power of a high-voltage transformer: available high-voltage wires and core sizes. Core shapes and winding pattern

show some, but not significant, effects on the achievable weight. See [36] for detailed analysis.

1) *High-Voltage Wires*: These are offered at limited sizes and voltage ratings, which restricts the design of a high-frequency high-voltage transformer. We survey 190 high-voltage wires (both single conductor wires and litz wires) from Teledyne Reynolds and Rubadue rated between 1 kV and 40 kV. For litz wire, we refer the “wire size” as the equivalent size of the litz bundle, and “the single-strand conductor size” as the litz wire size; for single-conductor high-voltage wires, we refer both terms as the size of the conductor.

1) *Wire size*: We find that the smallest off-the-shelf wire size increases with the wire rated voltage. Between 1 and 3 kV, the smallest wire is AWG 40; between 3 and 25 kV, it is around AWG 28; and above 25 kV, it is AWG 20. There is no available off-the-shelf wire at ≥ 40 kV. The average current-carrying capabilities of AWG 40, AWG 28, and AWG 20 wires are 25 mA, 0.6 A, and 3.9 A, respectively, at an average current density of $500 \text{ A}\cdot\text{cm}^{-2}$, a rule of thumb for designing transformers with reasonable copper losses. For an example high-voltage low-power transformer, e.g., outputting 10 kV and 500 W, using wires rated ≥ 10 kV (preferred for insulation purposes) would result in inefficient use of the wire; if we use wires rated at lower voltages, we can fully utilize the copper area, but face increased risk of arcing and engineering complexity (such as sectioning the windings).

2) *Single strand conductor size*: Similarly, we find that smallest single strand conductor sizes of high-voltage wires also increase with the rated voltage. For high-frequency designs, it is preferred to use a single strand conductor size at or smaller than the skin depth [56] because larger sizes increase copper losses owing to proximity effect, take up more window area, and add to the total weight. The higher the frequencies and the voltage, the fewer the wire options and, thus, more limitations on the designs of such a transformer.

3) *Insulation jacket*: We find that the insulation thicknesses, though generally increasing with voltage, vary even for wires rated at the same dc voltage from the same manufacturer. For example, 18-kV wires have an insulation thickness range between 0.3 and 0.65 mm, corresponding to an effective dielectric strength between 30 and $60 \text{ kV}\cdot\text{mm}^{-1}$.

In the first-generation transformer, the insulation of the secondary wire accounts for $\sim 15\%$ of the transformer weight. Thus, thinner jacket thickness is preferred.

4) *Effects on the weight*: All three limitations mentioned above are mismatched across available wires. For example, the wire with the smallest wire size does not have the smallest litz size or the thinnest jacket thickness. Therefore, the compounded inefficiencies of three factors may result in additional weight, and designing a high-voltage transformer with the most appropriate off-the-shelf wire or with customized wires (if possible) is preferred.

Fig. 12(a) shows transformer weights and Fig. 12(b) shows average current density of secondary wires (assuming uniform distribution and ignoring skin or proximity effect) when a transformer is designed using off-the-shelf wires or customized wires with two effective dielectric strengths (30 and $40 \text{ kV}\cdot\text{mm}^{-1}$). We

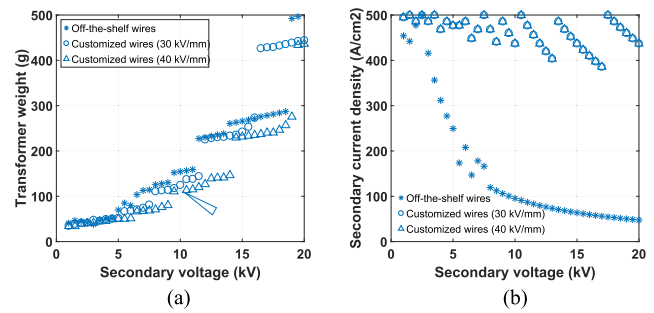


Fig. 12. (a) Weight and (b) secondary peak current density of high-voltage transformers designed with off-the-shelf wires or customized wires.

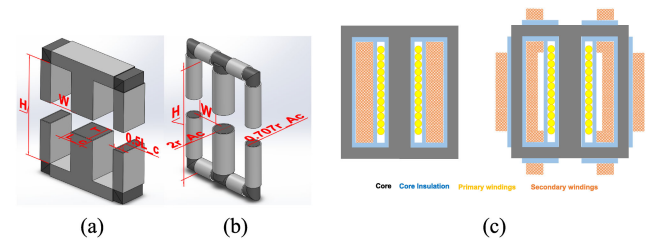


Fig. 13. (a) EE cores. (b) ER cores. (c) Front view of center-leg (left) and double-leg (right) winding considered in the weight study.

design the transformer to convert 500 V to 1 kV to 20 kV at 1 MHz and output 600 W. Four assumptions made across designs are as follows: 1) consider only Ferroxcube off-the-shelf core sizes; 2) use Ferroxcube 3F46 as the core material; 3) secondary wires are rated at six times the layer-to-layer secondary voltages; and 4) for customized wires, use AWG48 as both the minimal litz size and the minimal wire size.

We conclude from these results that when the secondary voltage is below 5 kV in amplitude, the weights are not significantly different by using either wire. As the secondary voltage increases, the transformer weight increases discretely—this is a compounded effect of both the wires and the core sizes. Comparing across different types of wires, off-the-shelf wires are used at a much lower current density, taking up the winding space, driving up the core sizes, and yielding heavier designs compared to the customized wires.

2) *Core Sizes, Shapes, and Winding Patterns*: Manufacturers offer limited and discrete core sizes, which fit general designs but may not be optimized for high-voltage designs.

EE [see Fig. 13(a)] and ER cores [see Fig. 13(b)] with customizable sizes are considered in the study. EE cores are easy to machine and customize; ER cores are favorable for high-voltage designs because they have fewer sharp corners. For both core shapes, we assume that each side leg has half the cross-sectional area of the center leg. For ER cores, we also compare center-leg winding and double-leg winding [see Fig. 13(c)].

Fig. 14(a) compares transformer weight when designed using off-the-shelf cores or customized cores. We hold the same design specifications and the assumptions as the study above. To separate the effect of the wires, we consider using customized wires with an effective dielectric strength of $30 \text{ kV}\cdot\text{mm}^{-1}$ across all designs (thus, the data with asterisks in Fig. 14 correspond to the data with circles in Fig. 12).

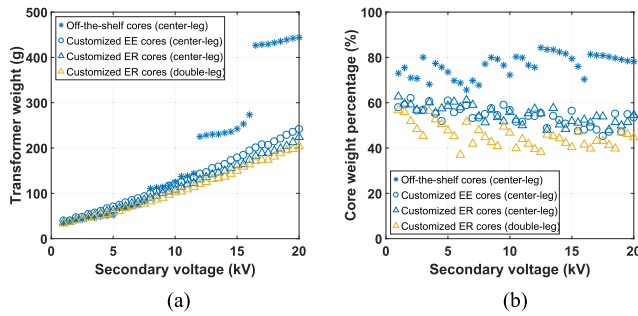


Fig. 14. (a) Transformer weight and (b) percentage of core weight in the transformer weight when designing high-voltage transformers with off-the-shelf cores or customized EE/ER cores.

As the secondary voltage increases (especially above ~ 10 kV), customized core sizes become more beneficial as there are fewer and more sparse off-the-shelf core sizes. Among the customized designs, as the voltage increases, the copper and the insulation takes up more of the transformer weight, and the core weight percentage reduces from 60% to 50% as the voltages increases from 0–5 to 15–20 kV [see Fig. 14(b)].

Fig. 14 also suggests that different core shapes and different winding patterns have some but not significant influence on the achievable transformer weight. With customization, a transformer weighing $10 \text{ g}\cdot\text{kV}^{-1}$ appears possible when stepping up a 500 V in amplitude 1 MHz ac to 5–20 kV at 600 W.

Furthermore, we can incorporate the improved weight study of voltage multipliers and high-voltage transformers to comprehensively design the three stages of the high-voltage dc–dc converter. The optimization methods are similar with that in Section III-C. With the improved approaches, higher specific power of the high-voltage dc–dc converter can be achieved [36].

VI. CONCLUSION

This article explores the design space and presents the design of a lightweight high-voltage converter for EAD propulsion applications. Various converter topologies are compared in terms of weight with considerations of device limitations. The weight of the converter is then minimized by design of the voltage gain of each stage. A prototype converter rated at 40 kV and 565 W is built and tested and achieves a specific power of $1.15 \text{ kW}\cdot\text{kg}^{-1}$, above other designs in its power and voltage class. Furthermore, several approaches to further improve the specific power of such a converter are presented. These design approaches can likewise be used to facilitate the miniaturization and weight reduction of high-voltage converters for many other applications.

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